

HealthConnect Clinic, Week 8

Final Analytics Report

Data Analytics Track: Final Integration, Presentation & Project Showcase

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1. Executive Summary

HealthConnect Clinic loses appointment slots to no-shows that are, in large part, predictable from data already on file at the time of booking. Across five weeks of analysis, testing and refinement, one factor stands out above the rest: how far in advance an appointment is booked. No-show risk rises steadily from 27.8% for bookings made within a week to 67.7% for bookings made 46 to 60 days out, and this pattern holds across every appointment type and age group tested.

A smaller but very high-risk group compounds this: patients booking 31 or more days out who also have two or more prior no-shows carry a 73.0% no-show rate. This is 233 of the dataset's 5,000 bookings, small enough to target directly, for example with a phone call instead of a standard reminder, and a pilot design for exactly that is included in this report.

Distance to the clinic matters, but only past a 15km threshold, not as a smooth gradient, so resourcing should follow that threshold rather than a general distance-based rule. Reminders help modestly, and cancellations increasingly give way to silent no-shows as lead time grows, suggesting that pairing long-lead reminders with an easy reschedule option is worth testing. Every finding in this report has been independently reproduced from the raw dataset, and one real discrepancy caught during testing, a wrong threshold in a cross-track model feature, was root-caused, corrected and re-validated under two different model types before being finalised here.

2. Final KPIs

KPI	Result
No-show rate by prior no-show bracket (KPI 1)	0: 43.5% 1: 53.5% 2: 59.4% 3+: 68.8%
No-show rate by booking lead time (KPI 2)	0-7d: 27.8% 8-14d: 33.6% 15-30d: 43.2% 31-45d: 53.8% 46-60d: 67.7%
No-show rate by distance to clinic (KPI 3)	<5km: 46.5% 5-10km: 46.5% 10-15km: 48.5% 15km+: 54.1%
No-show rate by reminder sent (KPI 4)	Reminder sent: 47.4% No reminder: 51.4%
Cancellation-to-no-show ratio (KPI 5)	Overall: 0.109 by lead band: 0.202 / 0.139 / 0.118 / 0.105 / 0.076

All five KPIs were independently recomputed directly from the raw prepared dataset (healthconnect_appointments_prepared.csv), bypassing the notebook that originally produced them, and matched to four decimal places. This confirms the dashboard reflects the underlying data and is not a notebook-only artefact.

3. Validated Findings

- **Booking lead time is the strongest predictor of no-show risk.** Confirmed across Week 5 point estimates, Week 6 bootstrap confidence intervals, Week 7 segment tests against appointment type and age group, and independently by the DS track collaborator's model, where it is the dominant feature.
- **The double-risk segment** (lead time 31+ days and 2+ prior no-shows) covers 4.66% of bookings at a 73.0% no-show rate, bootstrap 95% CI [67.4%, 79.0%]. The threshold was corrected from an initial 46+ days after a reproduction test failed, and the corrected 31+ definition was retested under two model types with no performance difference between thresholds. The 31+ definition was kept over 46+ days because it reaches nearly twice as many high-risk patients (233 vs. 123) at a still very high no-show rate (73.0% vs. 82.1%), so it flags more of the same validated risk pattern without materially diluting it.
- **Distance is a 15+ km threshold effect, not a gradient.** The three closer bands are statistically indistinguishable from each other. The 15+ km gap (about +7 points) holds inside every lead-time band and is not explained by lead time (correlation -0.008), but it is a modest effect on model performance (AUC +0.004).
- **Cancellations give way to silent no-shows as lead time grows.** The cancellation-to-no-show ratio falls from about 0.20 at 0-7 days to about 0.08 at 46-60 days, with non-overlapping confidence intervals at the two ends. This is a hypothesis worth testing with an intervention, not yet a proven causal result.
- **The reminder effect is modest** (47.4% no-show when a reminder is sent vs 51.4% when it is not) and was not tested to the same depth as lead time and distance this cycle.

4. Key Visualisations

The Power BI dashboard (outputs/dashboard/) presents four no-show rate charts with bootstrap confidence-interval error bars (prior no-show bracket, lead time, distance band, reminder sent), the KPI 5 cancellation-to-no-show ratio chart by lead-time band, and three tested slicers (appointment_type, age_group, reminder_sent).

5. Business Insights

- Booking lead time is the single most useful signal HealthConnect has for predicting no-shows before an appointment happens, and it is available at the moment of booking.
- A small group of patients (under 5% of bookings) accounts for a disproportionate share of high-confidence no-show risk, making them a realistic target for a higher-touch intervention rather than a blanket policy change.
- Distance is not a reason to treat all far patients the same way; only the 15+ km group shows a distinct, consistent effect.
- Long-lead-time bookings fail silently more often than they are actively cancelled, which existing cancellation-tracking processes will not catch.

6. Recommendations

- Use booking_lead_days as the primary variable for prioritising reminder effort and any future intervention work; treat it as a required, range-checked field in any downstream pipeline.
- Pilot a targeted phone call (rather than a standard reminder) for the double-risk segment (lead 31+ days, 2+ prior no-shows), following the design in Section 7.
- Apply a 15+ km threshold rule for distance-based resourcing rather than a graded distance policy.
- Test whether pairing long-lead-time reminders with an easy reschedule option reduces the silent no-show pattern; treat this as a hypothesis to validate, not a settled conclusion.

- Validate the reminder_sent and reminder_channel effects to the same depth as lead time and distance before acting on them further.

7. Double-Risk Call Pilot: Design

This cannot be run inside the current dataset, so it is presented as a design only. Items marked proposed need agreement with clinic staff before use.

Element	Design
Who is flagged	Lead time 31+ days and 2+ prior no-shows: 4.66% of bookings (233 of 5,000), baseline no-show rate 73.0%
Intervention (proposed)	A staff phone call in place of the standard SMS reminder
Comparison	Randomly split flagged patients into call vs standard reminder, same clinic and period
Primary metric	No-show rate. Secondary: cancellation/reschedule rate, staff minutes per call
Planning estimate	About 341 flagged patients per group (682 total) to detect a 10-point drop at 5% significance, 80% power; roughly 14,600 bookings at the 4.66% flag rate
Success rule (proposed)	A drop of at least 10 points, with the 95% interval excluding zero
Status	Design only. Nothing has been run and no pilot results are claimed.

8. Analytical Limitations

- All findings are validated on a single fictional dataset of 5,000 bookings; all model checks are cross-validation on the same rows, with no held-out or external data. AUC values may be optimistic and should be described as validated on this dataset only.
- 90 of 5,000 rows (1.8%) have no distance recorded and are excluded from distance tests; unlikely to change the direction of the finding.
- The 3+ prior no-show bracket in KPI 1 is small with a wide interval and should be treated as fragile.
- Distance groups inside individual lead-time bands are small (118 to 259 patients at 15+ km); results use simple intervals rather than a bootstrap, and distance has not been tested against prior no-shows.
- reminder_sent and reminder_channel effects have not been tested to the same depth as lead time and distance and remain a lower-confidence finding.
- The double-risk call pilot is a design only; no real-world evidence yet exists that the intervention reduces no-shows.

9. Cross-Track Contribution

The Data Analytics track's cross-track exchange this cycle was with the DS track collaborator: a validated correction to the double-risk segment threshold (31+ days rather than 46+ days), reproduced directly from the shared prepared dataset and retested under both a Random Forest and the exact Week 6 candidate Logistic Regression, with no performance difference found between thresholds. The full HC-POD integration record, covering dependencies, outputs exchanged and evidence, is documented separately in the Week 8 HC-POD Integration Record.